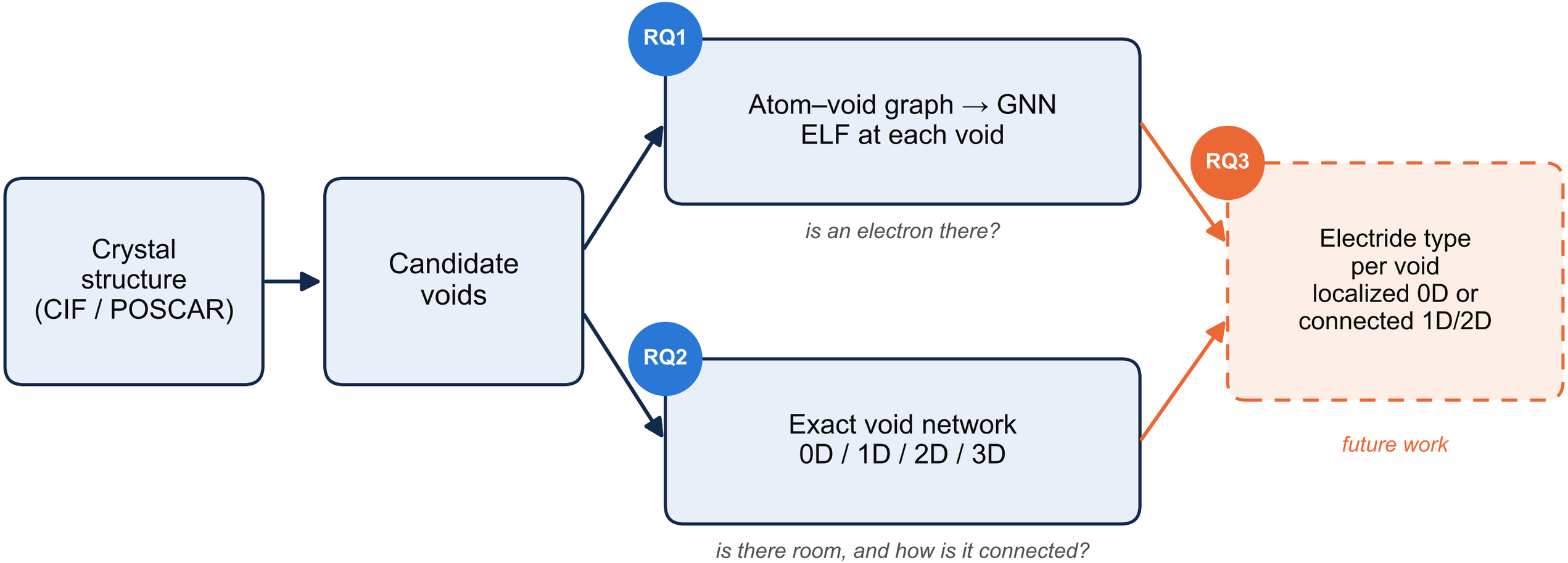
Crystal structure (CIF / POSCAR)
Candidate voids
RQ1
Atom–void graph → GNN
ELF at each void
is an electron there?
RQ2
Exact void network
0D / 1D / 2D / 3D
is there room, and how is it connected?
RQ3
Electride type per void
localized 0D or
connected 1D/2D
future work